

Synopsis

On

Spatial-AI Digital Twin for Autonomous Warehouses



Group ID: TY-AIA409

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1. Title of the Project

Spatial-AI Digital Twin for Autonomous Warehouses

The title is intentionally concise while representing the project's primary technologies: Spatial AI, Digital Twins, autonomous warehouse systems, real-time intelligence, and fleet management.

The synopsis template specifically asks for a concise and descriptive project title reflecting the project's purpose.

2. Abstract

Modern autonomous warehouses depend on Automated Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs) to perform material handling and fulfillment operations. However, conventional warehouse monitoring systems are largely passive, relying on historical telemetry, threshold-based alerts, and static visualization. These systems may fail to identify physical anomalies such as wheel slippage, abnormal temperature rise, actuator friction, or environmental hazards before they affect operations.

This project proposes a Spatial-AI Digital Twin that combines real-time telemetry, Physics-Informed Neural Networks (PINNs), edge-based inference, and intelligent fleet rerouting. A synthetic telemetry pipeline will generate realistic robot states and controlled anomaly conditions to address the limited availability of labeled industrial failure data. The PINN model will learn expected physical behavior and identify deviations between observed and physics-consistent states. Detected anomalies will be converted into risk information and used by a dynamic routing engine to generate safer alternative paths. A spatial digital twin dashboard will visualize robot positions, operational states, anomalies, risk zones, and routing decisions. The proposed system aims to enable predictive anomaly detection, faster response, improved fleet safety, reduced downtime, and more efficient warehouse operations.

3. Introduction / Background

Modern warehouses are increasingly adopting AMRs, AGVs, robotic handling systems, sensors, and automated fleet-management platforms to improve operational efficiency. These systems continuously generate telemetry such as position, velocity, temperature, battery condition, and motion-related parameters.

Digital Twin technology provides a virtual representation of physical assets and environments. However, many existing implementations primarily function as monitoring and visualization systems, displaying the current or historical state of physical assets rather than predicting how those assets may behave under abnormal conditions.

A major challenge is that real industrial anomaly datasets are difficult to obtain because they involve proprietary infrastructure, safety concerns, operational restrictions, and limited failure events. Conventional machine-learning approaches may therefore require large quantities of labeled data that are difficult to acquire.

The proposed project addresses this gap by combining:

- Spatial Digital Twins
- Physics-Informed Neural Networks
- Synthetic telemetry generation
- Edge AI
- Real-time anomaly detection
- Dynamic fleet rerouting

Instead of treating the digital twin as only a visualization layer, the system aims to make it an active predictive and decision-support layer.

4. Problem Definition

Modern autonomous warehouses rely on AMRs and AGVs, but lack real-time Spatial-AI, Physics-Informed anomaly prediction, and intelligent fleet rerouting, leading to delayed hazard detection, safety risks, asset damage, and operational downtime.

Existing approaches generally depend on:

- Passive digital-twin visualization
- Historical telemetry
- Fixed threshold-based alerts
- Static or rule-based routing
- Centralized intervention
- Large labeled datasets for conventional ML

These limitations reduce the ability to understand physical behavior and identify previously unseen anomalies in real time. The proposed project therefore focuses on connecting the physical or simulated warehouse state to AI prediction, spatial representation, risk assessment, and autonomous routing decisions.

5. Objectives

1. To develop a synthetic telemetry pipeline for generating realistic AMR/AGV operational data.
2. To implement a Physics-Informed Neural Network (PINN) for identifying deviations from expected physical behavior.
3. To detect physical anomalies such as wheel slippage, overheating, and abnormal mechanical behavior.
4. To develop an edge-based inference pipeline capable of performing low-latency anomaly evaluation.
5. To implement dynamic fleet rerouting when a hazardous or high-risk region is detected.
6. To develop a Spatial Digital Twin dashboard for visualizing warehouse assets, robot states, anomalies, and risk zones.
7. To evaluate the system based on anomaly-detection performance, inference latency, routing response, and operational visualization.

6. Scope of the Project

The project will focus on a simulated autonomous warehouse environment containing AMRs/AGVs and associated telemetry.

Included Scope

- Synthetic AMR/AGV telemetry generation
- Robot position and velocity tracking
- Battery and temperature monitoring
- Slip/anomaly simulation
- Physics-informed anomaly detection
- Risk assessment
- Edge-based inference
- Dynamic fleet rerouting
- Spatial Digital Twin visualization
- Real-time dashboard
- Performance evaluation

Out of Scope

- Physical deployment on industrial AMRs
- Real warehouse control-system integration
- Safety certification for industrial deployment
- Direct control of physical robots
- Large-scale commercial warehouse deployment

The project will therefore serve as a functional research prototype and proof of concept.

7. Proposed System

7.1 Overall Concept

The proposed system transforms the conventional Digital Twin from a passive visualization layer into an active predictive intelligence layer.

AMR/AGV Telemetry → Data Processing → Physics-Informed AI → Anomaly Detection → Risk Assessment → Dynamic Rerouting → Spatial Digital Twin → Operator

7.2 Major Modules

Module 1 — Telemetry & Data Engineering

Generates and processes high-frequency robot telemetry such as:

- Position

- Velocity
- Acceleration
- Battery level
- Battery temperature
- Wheel/slip ratio
- Motor-related parameters
- Environmental conditions

Controlled anomalies can be injected into the simulated data to create realistic test scenarios.

Module 2 — Physics-Informed AI

A Physics-Informed Neural Network (PINN) will learn the expected physical behavior of the robot.

The model combines:

$\text{Data Loss} + \text{Physics Loss} = \text{Total Training Loss}$

Physics constraints can represent relevant relationships involving:

- Motion
- Velocity
- Acceleration
- Energy/thermal behavior
- Physical state consistency

The model will calculate deviations between the observed physical state and the expected physics-consistent state.

Large deviations can indicate potential anomalies.

Module 3 — Edge Intelligence & Fleet Rerouting

The anomaly prediction module produces a risk score for the robot or environment.

When a high-risk condition is detected:

Anomaly → Risk Zone → Route Update → Collision-Free Alternate Path

The system will attempt to perform inference and routing with low latency, with sub-50 ms as the project target/benchmark, rather than claiming it as an already achieved result.

Module 4 — Spatial Digital Twin

The dashboard provides a spatial representation of the warehouse.

It will display:

- Warehouse layout
- AMR/AGV locations
- Robot movement
- Robot health
- Anomaly locations
- Risk zones
- Current routes
- Alternate routes
- Alerts

This creates a visual connection between the physical/simulated environment and its digital representation.

The synopsis specifically recommends describing the architecture/workflow, modules, functions, advantages, and optionally including a block diagram.

8. System Requirements

A. Hardware Requirements

Development System

- Processor: Intel Core i5 / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB; 16 GB recommended
- GPU: NVIDIA GPU with CUDA support preferred
- Storage: Minimum 20 GB available
- Display: Full-HD recommended
- Internet: Required for software/package installation

Edge Hardware — Optional

For future edge deployment:

- NVIDIA Jetson platform
- CUDA-enabled edge device
- Suitable sensor/telemetry interface

The edge hardware is a deployment target, not necessarily a mandatory requirement for the initial prototype.

B. Software Requirements

Category	Technology
Programming Language	Python

Category	Technology
AI/ML	PyTorch
Numerical Computing	NumPy, SciPy
Data Processing	Pandas
Visualization	Matplotlib / Plotly
Digital Twin Interface	Streamlit
3D Visualization	PyVista / Three.js
Development	VS Code / Jupyter Notebook
Version Control	Git / GitHub
Edge Acceleration	CUDA / NVIDIA Jetson
Data Storage	CSV / SQLite / PostgreSQL
Routing	Graph-based path planning

9. Methodology & Implementation Plan

Development Methodology

Agile / Iterative Development

The project will be developed incrementally through independent modules. Each module will be tested before integration with the complete system.

Implementation Phases

Phase 1 — Requirement Analysis

- Identify warehouse use case
- Identify robot parameters
- Define anomaly scenarios
- Define system architecture
- Identify performance metrics

Phase 2 — Telemetry Simulation

- Generate synthetic AMR/AGV data

- Model normal robot behavior
- Introduce controlled anomalies
- Validate telemetry consistency

Phase 3 — Data Processing

- Data cleaning
- Normalization
- Feature engineering
- Time-series preparation
- Train/test data separation

Phase 4 — PINN Development

- Define neural-network architecture
- Define physics constraints
- Develop physics-informed loss
- Train model
- Validate anomaly prediction

Phase 5 — Anomaly & Risk Engine

- Calculate prediction residuals
- Establish adaptive anomaly thresholds
- Generate anomaly scores
- Classify risk levels

Phase 6 — Fleet Rerouting

- Represent warehouse as a navigation graph
- Identify blocked/high-risk regions
- Generate alternate routes
- Evaluate route safety and efficiency

Phase 7 — Spatial Digital Twin

- Create warehouse environment
- Add robot visualization
- Display telemetry
- Display anomaly zones
- Display routes and alerts

Phase 8 — Edge Optimization

- Optimize inference pipeline
- Measure latency
- Test lightweight deployment
- Evaluate target sub-50 ms response

Phase 9 — Integration & Testing

- Integrate all four modules
- Perform anomaly scenarios
- Test rerouting
- Test dashboard
- Measure system performance

Algorithms & Techniques

Core AI

- Physics-Informed Neural Networks (PINNs)
- Residual-based anomaly detection
- Statistical thresholding
- Risk scoring

Data Engineering

- Synthetic data generation
- Feature engineering
- Data normalization
- Time-series processing

Routing

- Graph-based path planning
- Dynamic route updating
- Collision/risk-aware path selection

Visualization

- Spatial Digital Twin
- 3D warehouse representation
- Real-time telemetry visualization

10. Expected Outcome

The expected outcome is a functional prototype of an AI-enabled Spatial Digital Twin capable of:

- Generating realistic AMR/AGV telemetry
- Detecting simulated physical anomalies
- Identifying deviations using physics-informed intelligence
- Producing real-time anomaly/risk information
- Dynamically rerouting affected robots

- Visualizing robot states and risk zones
- Providing real-time operator alerts
- Demonstrating low-latency edge inference
- Reducing dependency on large labeled failure datasets

Key Evaluation Metrics

Metric	Purpose
Anomaly Detection Accuracy	Measures detection capability
Precision / Recall / F1	Measures classification quality
False Positive Rate	Measures unnecessary alerts
Inference Latency	Measures real-time performance
Route Planning Time	Measures rerouting responsiveness
Path Efficiency	Compares route quality
System Availability	Evaluates operational reliability

11. Applications / Use Cases

Autonomous Warehouses: Real-time monitoring and routing of AMRs/AGVs in fulfillment centers.

Smart Manufacturing: Monitoring autonomous vehicles and robotic systems in industrial facilities.

Predictive Maintenance: Early identification of abnormal equipment behavior before major failures.

Industrial Safety: Detection and visualization of hazardous operational conditions.

Fleet Management: Dynamic coordination and rerouting of multiple autonomous robots.

Logistics & Distribution Centers: Improving material movement and reducing operational bottlenecks.

Industrial Digital Twins: Using AI-enabled digital twins for predictive monitoring and decision support.

12. Future Enhancements / Future Scope

Autonomous Warehouses: Real-time monitoring and routing of AMRs/AGVs in fulfillment centers.

Smart Manufacturing: Monitoring autonomous vehicles and robotic systems in industrial facilities.

Predictive Maintenance: Early identification of abnormal equipment behavior before major failures.

Industrial Safety: Detection and visualization of hazardous operational conditions.

Fleet Management: Dynamic coordination and rerouting of multiple autonomous robots.

Logistics & Distribution Centers: Improving material movement and reducing operational bottlenecks.

Industrial Digital Twins: Using AI-enabled digital twins for predictive monitoring and decision support.

13. References

1. Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707.
2. Kritzinger, W., Karner, M., Traar, G., Henjes, J., & Sihn, W. (2018). Digital Twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11), 1016–1022.
3. Tao, F., Zhang, H., Liu, A., & Nee, A. Y. C. (2019). Digital Twin in Industry: State-of-the-Art. *IEEE Transactions on Industrial Informatics*, 15(4), 2405–2415.
4. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
5. PyTorch Documentation — Deep learning framework used for implementing the PINN architecture.
6. NumPy & SciPy Documentation — Numerical computation and telemetry simulation.
7. Streamlit Documentation — Interactive dashboard development.
8. PyVista Documentation — 3D scientific visualization.
9. NVIDIA Jetson Documentation — Reference for future edge-AI deployment.